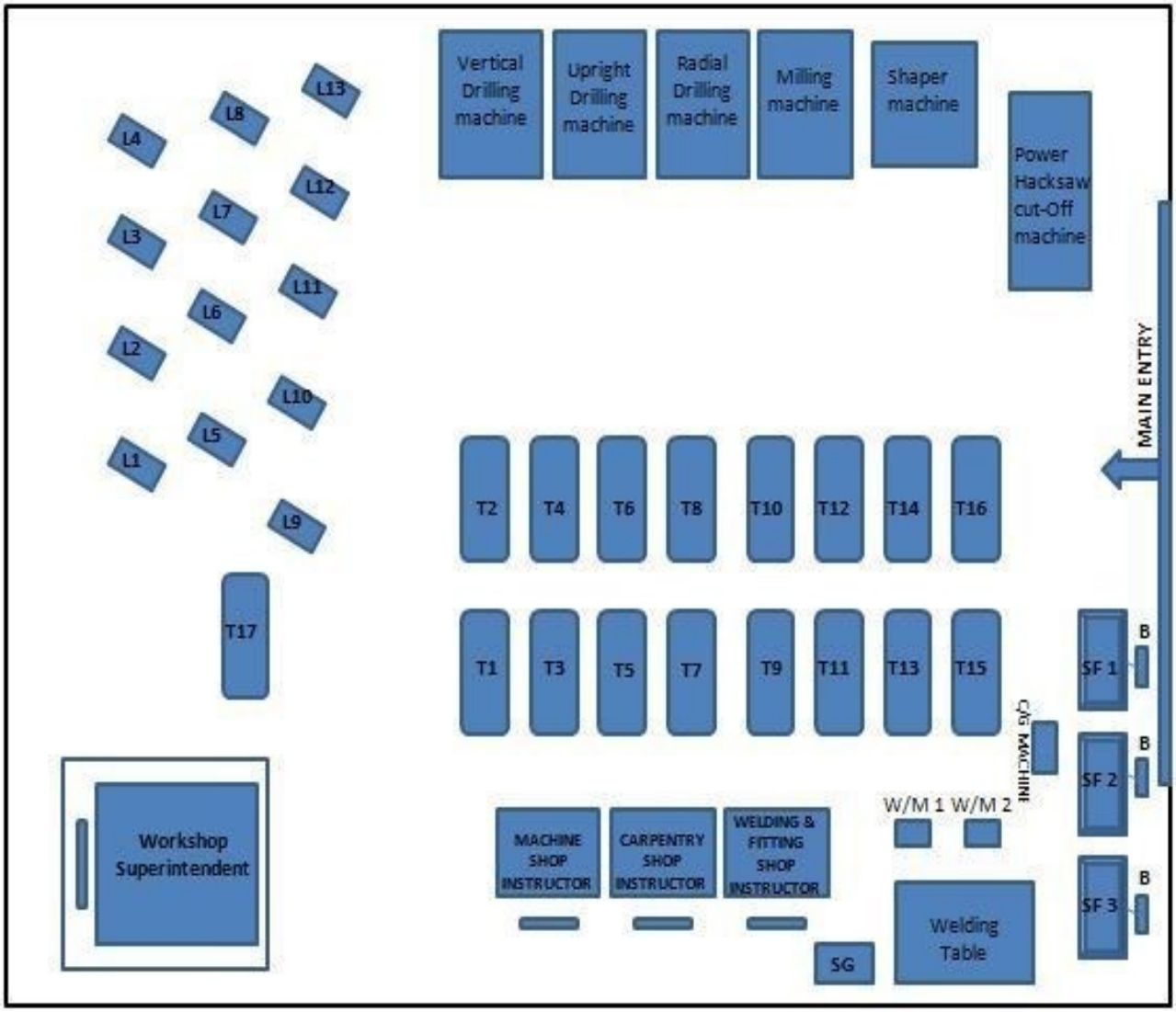
Practical No. 0

Workshop Layout Diagram:



L- Lathe machine
T- Table
SG – Surface Grinder
W/M- Welding Machine
SF- Smithy Shop Furnace
B- Blower
C/G – Metal Cut - Off Grinder Machine

Aim: Know your lab (Mechanical Workshop).

College Vision & Mission:

Vision: To be a center of excellence imparting professional education satisfying societal and global needs.

Mission: Transforming Students into lifelong learners through quality teaching, training and exposure to concurrent technologies fostering conductive atmosphere for research and development through well equipped laboratories and qualified personnel in collaboration with global organizations.

The JDCOEM is adhere to high quality parameters to ensure the effective outcome of various academic, curricular and co-curricular interventions on learner centric environment through:

- Domain knowledge
- Strong Industry
- Life skills
- Lifelong learning
- Department vision and mission

Vision: To lay a robust foundation for the institute to reach its zenith.

Mission: The department is making its paramount effort

Achieving academic excellence through rigorous teaching, learning and evaluation practices.

To develop an ability to apply knowledge of basic science and mathematics to excel in the field of engineering.

To provide salutary environment for the betterment of faculties & students.

Salient Feature of Department:

- Departmental library
- Modern laboratory
- Separate tutorial rooms

- Well trained and qualified supporting staff
- Modern and spacious Drawing hall

Staff of Workshop:

Workshop Superintendent: Prof. Rakesh Bandane

Machine Shop Instructor: Mr. Gopal Domke

Carpentry Shop Instructor: Mr. Sheshrao Sontakke

Welding / Fitting Shop Instructor: Mr. Shashikant Rameke

Workshop objectives:

Know basics of workshop processes.

Read and interpret job drawing.

Identify, Select and cutting tools and equipments.

Inspect job for specified dimensions.

Produce jobs as per specified dimensions.

Carpentry Shop:

Carpentry may be defined as the process of making wooden components. It is starts from a marketable form of wood s and ends with the finished product. It deals with the building work, furniture, cabinet making etc.

Tools for the carpentry shop:-

Wooden mallet, claw hammer, pincer, try square, hand saw , marking gauge, chisels, measuring tape, chalk lines, wood rasp file, jackplane, smooth plane,carpentry vice , screw , axe.

Application of carpentry Shop:

Following blueprints and building plans.

Installing structures and fixtures.

Measuring, cutting and shaping wood, plastic and other material.

Fitting Shop:

In fitting shop unwanted material is removed with the help of hand tools. It is done for mating, repair and manufacturing purpose. Commonly used tools in fitting shops are hacksaw frame with blade, file and center punch etc.

Tools for the fitting shop:

Clamping tools bench vice, leg vice, cutting tools, vernier height gauge, inside caliper, divider, surface plate, steel rule, measuring tape.

Application of fitting shop:

The hand operations in fitting shop include marking, filing, sawing, scraping, drilling etc.

These are performed using hand tools or power operated portable tools.

Solar panel support and racking system.

Graphics Display support structures.

Welding Shop:

Welding is a fabrication process whereby two or more parts are fused together by means of heat, pressure or both forming a join as the parts cool. Welding is usually used on metals.

Or

A Weld can be defined as coalescence of metals produced by heating to a suitable temperature with or without the application of pressure and with or without the use of a filler material.

Tools for the welding shop:

Safety glasses, Auto darkening welding helmet, chipping hammer, wire brush, welding gloves, torch tip, and cleaner.

Application of Welding Shop:

- Resistance Welding: Automobile Parts
- Thermite Welding: Rail Joints in railway
- Tungsten Inert Gas Welding: Aerospace and nuclear reactors.
- Submerged Arc Welding: Heavy Engineering Ship Building
- Gas Metal Arc Welding: Joining of Metals

Some instructions for students when to do the practicals:

Carpentry:-

[1] Be aware of follow all municipal provincial and federal codes, Regulations.
[2] Inspect all equipments before use.
[3] Keep all equipments in good & repaired condition.
[4] Wear apron while doing practical.

Fitting Shop:-

[1] Keep hands and tools wiped clean and free of dust, oil and grease. Dry tools are are safer to use.
[2] Do not carry sharp tools in packets.
[3] Were safety/leather/sport shoes, not sandles/chappal.
[4] Wear apron while doing practical.

Welding Shop:-

[1] Use black glass shield helmets while doing practical.
[2] Were safety/leather/sport shoes, not sandles/chappal.
[3] Always were a long sleeved, non-flammable clothes.
[4] Use hand gloves while doing welding operation.